Consumer/Distributor level food and fuel adulteration detection tools using FT-IR spectroscopy and multivariate data analytics

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Abstract

Food and fuel adulteration are widely common in countries like India, usually due to profitability reasons and sometimes due to the scantiness of the resource. Such adulterations are happening both at production and distribution levels. And there are no successful tools available at consumer and distributor levels which can check for adulteration. FT-IR spectroscopy along with multivariate data analytics is one of the most commonly used methodologies in food and fuel adulteration detection tools. In this paper I would like to present a startup idea which uses FT-IR principles along with chemo-metric analytics to create a commercial tool which can be used by both consumers and distributors to check for adulteration with a relative ease of use.

1. Introduction

Fuel and food adulteration is one of the major concerns in developing countries like India. Adulteration of fuel in-turn is one of the important causes of air pollution and engine related problems in the vehicles. The reason for this kind of adulteration is due to the differential taxation present in most of the developing countries. In India gasoline is taxed more than diesel and diesel is taxed more than kerosene. So there is a common practice of mixing cheaper hydro-carbon fuels with highly taxed hydrocarbon, for the profitability reasons. These adulterations happen at both production and distribution levels. In a survey conducted by Tata consultancy services it was found that more than 30% of the kerosene intended for household consumption flowed back to the industry in one form or the other[1]. Usually petrol is found to be adulterated using naphtha and kerosene, whereas diesel is found to be adulterated with kerosene.

Milk, yogurt and edible oils are the most commonly adulterated foods in most of the Asian countries like India and China. As per the survey conducted by Food safety and standards Association of India (FSSAI) 68% of country's milk doesn't conform to the standards set [2]. Milk scandal (2008) which occurred in china affected at least 300,000 people out of which 54,000 are babies [3]. Buying and selling of loose edible oils is very common in India. This is mostly because of varied economic conditions and availability of options. Most common adulterants mixed with milk are usually water, glucose and skim milk powder, detergent, fat and urea. With respect to edible oil the most common additive is Argemone oil, which is non-edible and harmful.

Many techniques have been developed for adulteration checks both in food and fuel industries. Most of these methods exploit the fact that the molecules absorb at specific frequencies that are characteristic of their structure [4], which in-turn helps in identifying the presence of different molecules in any given sample. Absorption spectroscopy measures the absorption of radiation as a function of wavelength or frequency when radiation interacts with a sample [4]. These techniques are used in analytical chemistry to look for the presence of particular substance in a sample. The same techniques are used in adulteration detection kits to look for the presence of adulterants. Most of the commercial tools available in developed countries for fuel adulteration checks use the combination of Fourier transform Infrared spectroscopy (FT-IR) Imaging techniques with chemo-metric analytics. In the case of food there are wide varieties of techniques available. Table.1 gives an overview of techniques and their applications. Among the methods described in the table.1 IR spectroscopy based methods are widely used in adulteration detection of milk and edible oils.

Till date no successful consumer and distributor level adulteration detection tools which are cost and technically effective are available in India. In this paper I would like to present a startup idea which can make use of the need for such a multi-purpose adulteration detection tool. The proposed tool utilizes the principles of

Infrared spectroscopy combined with multivariate data analytics.

Techniques	Applications
IR Spectroscopy	Fruit juice; wine; honey
Raman Spectroscopy	Olive oil; honey
Thermal Analysis	Palm oil, honey
Enzyme-linked Immuno Sorbent Assay (ELISA)	Meat; milk

Table.1 Techniques and their applications [5]

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2. Technical Rationale

Infrared spectroscopy is a widely used methodology for quantification of sample, both in organic and in-organic industries [4]. Absorption of infrared radiation causes change in the molecular vibrations within molecules. A sample can be analyzed by passing infrared radiation over a wide range of frequencies and absorptions made by each type of bond in the sample can be measured. This is usually plotted as percentage transmittance versus wavenumber and the Fourier transform of this plot will give the infrared spectrum of the sample. As no two compounds have same infrared spectrum, adulterated samples can be identified by comparing them with spectra of pure samples [4].

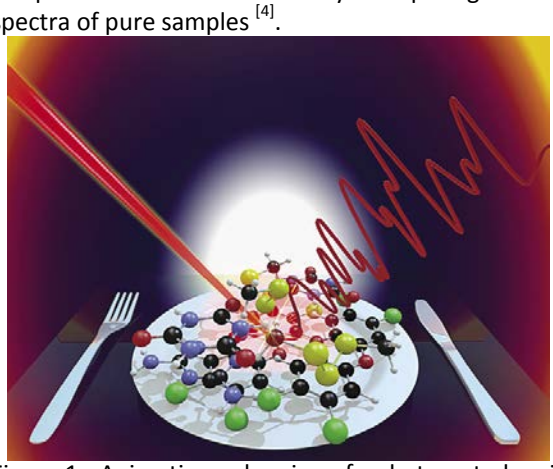

Figure.1 Animation showing food targeted with IR radiation and resultant spectrum [6].

So the IR spectrum of each pure sample, be it milk, edible oil, petrol and diesel can be taken as a reference fingerprint and can be compared against the adulterated versions using chemo-metric analysis

2.1. Method

The methodology for the adulteration detection can be divided into two stages

Sample preparation

Different samples of milk, edible oil, petrol and diesel, with no adulterants and with adulterants of known amounts should be prepared.

Obtaining IR spectral Images, Feature computation, training and classification

Using a FTIR spectrometer IR spectrum of each sample is to be obtained. A typical IR spectrum is shown in figure.2. The data used for the spectral plot shown is obtained from university of Manchester [7].

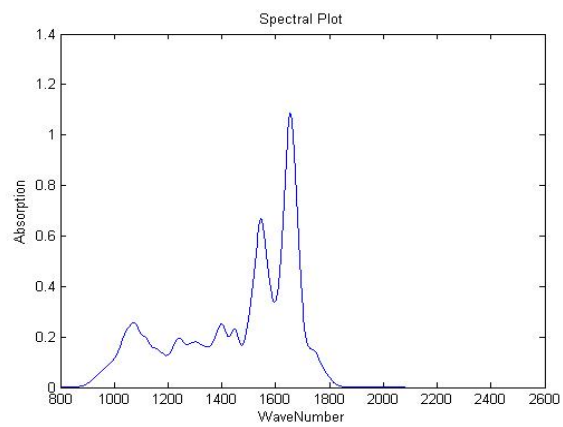

Figure.2 Spectral plot of a sample

Usually spectral data consists of two columns, one is the wavenumber column and the other is the absorption column.

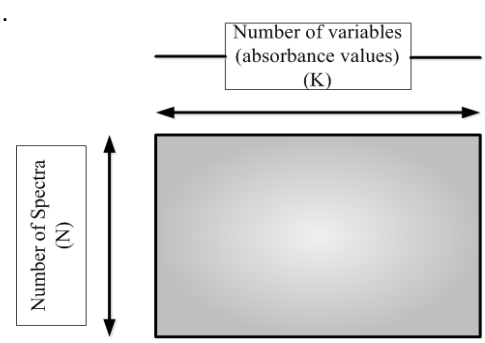

Figure.3 Absorption Matrix (N X K) [7]

All the absorption value columns of different samples are appended and an absorption matrix is formed. An absorption matrix typically looks like as shown in the figure 3, each row corresponds to spectrum of a different sample. And each column corresponds to number of data points in each spectrum. A similar absorption matrix can be formed for pure milk samples, and milk samples adulterated with different additives and for other classes like edible oil, petrol and diesel the same methodology can be repeated. After absorption matrix is formed the data is to be preprocessed using some dimensionality reduction methods like PCA to remove redundancy in the data. After preprocessing the absorption matrix can be divided into training, validation and test sets to train and test the classifier. There are many classification algorithms usually used in classification of adulterated food and fuels. TABLE 2 represents common classification algorithms used and their methods [6]. For the present algorithm, principal component analysis (PCA) is used both for feature reduction and detection of boundaries for each class.

Modeling Method	Model Type	Supervised/ Un Supervised
PCA	Explanatory	Un Supervised
Self-organizing maps	Explanatory	Un Supervised
Partial least square regression	Quantitative	Supervised
Discriminant Analysis	Classification	Supervised

Table.2 Chemo-metric modeling techniques [6]

2.1 Example Results

The hypothetical results which can be obtained using FT-IR spectral images and PCA are shown for milk and petrol in figures 4 and 5.

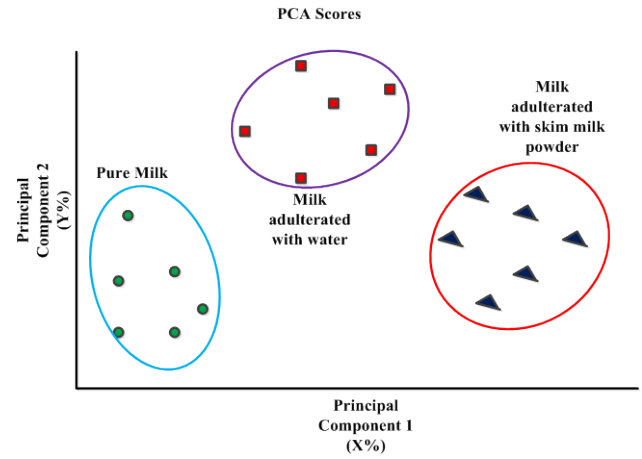

Figure.4 Classes Identification for Milk samples

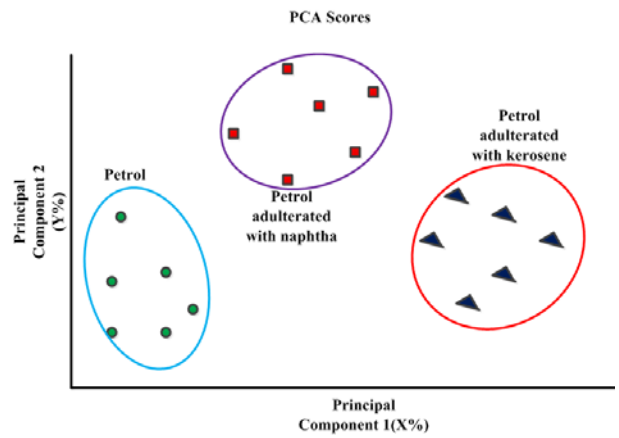

Figure.5 Classes Identification for Petrol samples

Leonardo et.al [8], Downey et.al [9], He et.al [10], used combination of FT-IR spectral data along with different data analytic techniques and could successfully achieve their objective of adulteration detection and quantification of adulterants.

2.2. Proposal for the start-up

Goal: A cost effective, portable Food/Fuel adulteration detection toolkit targeting both distributor and consumer level base in India. A hypothetical figure of such a toolkit would look like as shown in figure.6.

Spectrometer Design:
Portable FourierTransform Infrared spectrometer (PFT-IR)

Programmable Software Design:
Features: FT-IR spectral content.
Feature reduction: Principal component Analysis
Sample classification: Principal component Analysis
Quantitative adulterant measurement: Partial Least Squares

Commercial Food and Fuel adulteration detection kit

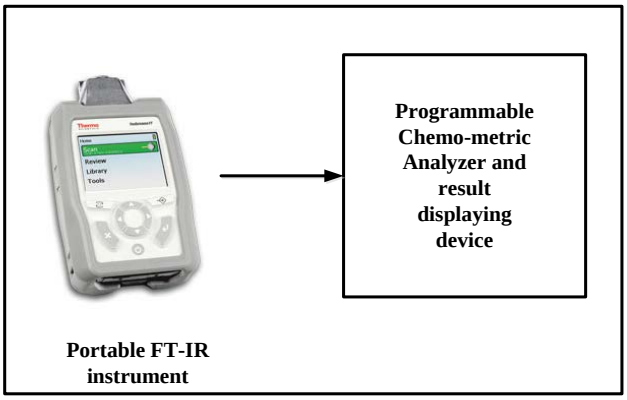

Figure.6 Proposed tool-kit

2.3. Technical Challenges/Risks

a) Portable FTIR spectrometers are usually costly and there are no indigenous companies in India which manufacture such items. Importing them from foreign countries will increase the cost of the units. As the fuel quality changes between metropolitan cities and the others a varied design of the software is needed.

b) Samples with mono-atomic ions and homo nuclear molecules don't have IR spectra and are difficult to analyze. Complex mixtures and aqueous solutions are difficult to analyze.

c) Usage of harmful chemicals / Machinery is not involved so there are no technical risks involved

2.4. Commercial Rationale and Viability

Commercial Rationale:

a) No consumer or distributor level tools are available to check for adulteration.

b) Few tools like ELISA which requires the use of antibodies to look for adulteration in the sample require domain knowledge and are limited to few known adulterants.

c) Lactometer based approaches are not accurate and reliable.

d) There is a high need of such tool as the adulteration is happening both at production and distribution level.

Commercial Viability:

a) Project has a high potential business value

b) No such tool which is cost and technically effective at consumer and distributor level is available

c) Since device is portable and cost effective, longer period of demand can be expected.

d) Distributor level kits can easily be commercialized

e) Development of cost effective indigenous portable FT-IR spectrometers can make targeting consumer level easier.

2.5. Commercial Risks

As the proposed software is programmable, it is prone to fraudulent acts like rigging. Such acts might hurt the belief of consumers and distributors which in-turn affects the business.

3. Discussion and Conclusion

FT-IR principles along with data analytics proved to be a most successful way of detecting adulteration in wide variety of foods and fuels. The proposed tool-kit uses FT-IR spectral content, Principal Component Analysis and Partial Least Squares for adulteration detection and content quantification of samples. Present target of the company would be the consumer base and Food/Fuel distributors in India

Portable IR spectrometers are costly and are not readily available in India. Targeting a distributor base can be readily done, whereas more efforts are required to make the tool commercially successful at consumer level. Encouraging indigenous markets which can manufacture cheap and portable FT-IR spectrometers will make targeting consumer base easier.

4. References

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